

SECTION 220523 - GENERAL-DUTY VALVES FOR PLUMBING PIPING

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

- A. Drawings and general provisions of the Contract, including General and Supplementary Conditions and Division 01 Specification Sections, apply to this Section.

1.2 SUMMARY

- A. Section Includes:
 - 1. Bronze angle valves.
 - 2. Bronze ball valves.
 - 3. Iron, single-flange butterfly valves.
 - 4. Iron, grooved-end butterfly valves.
 - 5. Bronze swing check valves.
 - 6. Iron swing check valves.
 - 7. Iron, grooved-end swing-check valves.
 - 8. Bronze gate valves.
 - 9. Iron Gate valves.
 - 10. Bronze globe valves.
 - 11. Iron globe valves.
- B. Related Sections:
 - 1. Division 22 plumbing piping Sections for specialty valves applicable to those Sections only.
 - 2. Division 22 Section "Identification for Plumbing Piping and Equipment" for valve tags and schedules.

1.3 DEFINITIONS

- A. CWP: Cold working pressure.
- B. EPDM: Ethylene propylene copolymer rubber.
- C. NBR: Acrylonitrile-butadiene, Buna-N, or nitrile rubber.
- D. NRS: Nonrising stem.
- E. OS&Y: Outside screw and yoke.
- F. RS: Rising stem.

1.4 SUBMITTALS

- A. Product Data: For each type of valve indicated.

1.5 QUALITY ASSURANCE

- A. Source Limitations for Valves: Obtain each type of valve from single source from single manufacturer.
- B. **ASME Compliance:**
 - 1. ASME B16.10 and ASME B16.34 for ferrous valve dimensions and design criteria.
 - 2. ASME B31.1 for power piping valves.
 - 3. ASME B31.9 for building services piping valves.
- C. NSF Compliance: NSF 61 for valve materials for potable-water service.

1.6 DELIVERY, STORAGE, AND HANDLING

- A. Prepare valves for shipping as follows:
 - 1. Protect internal parts against rust and corrosion.
 - 2. Protect threads, flange faces, grooves, and weld ends.
 - 3. Set angle, gate, and globe valves closed to prevent rattling.
 - 4. Set ball and plug valves open to minimize exposure of functional surfaces.
 - 5. Set butterfly valves closed or slightly open.
 - 6. Block check valves in either closed or open position.
- B. Use the following precautions during storage:
 - 1. Maintain valve end protection.
 - 2. Store valves indoors and maintain at higher than ambient dew point temperature. If outdoor storage is necessary, store valves off the ground in watertight enclosures.
- C. Use sling to handle large valves; rig sling to avoid damage to exposed parts. Do not use hand wheels or stems as lifting or rigging points.

PART 2 - PRODUCTS

2.1 GENERAL REQUIREMENTS FOR VALVES

- A. Refer to valve schedule articles for applications of valves.
- B. Provide angle valves for incoming water supply to all fittings.
- C. Gate valves up to 50mm diameter shall be bronze/DZR body with threaded ends bronze wedge disc, high quality packing and gland, non-rising stem with screwed bonnet.
- D. Check valves up to 50mm diameter shall be threaded bronze pattern swing type with renewable disc and screw-in cap.
- E. Valve Pressure and Temperature Ratings: Not less than indicated and as required for system pressures and temperatures.
 - 1. Unless otherwise indicated, valves on systems with working pressure up to 12bar shall be rated for 16 Bar minimum and as required.

2. **Gate Valve – Flanged (65 and 300 mm):** shall be in accordance with DIN 8077 & 8078(PN 20) and fittings in accordance to DIN 16962, (PN25) Standard and PPR composite, green color, and Pressure Rating, PN16 or PN25, as per system pressure requirements, Operation: Hand Wheel.
 3. **Ball Valve -- Threaded (lever operated) (15 mm to 50 mm):** Bronze or DZR copper alloy body, Threaded, to BS- 21, Chrome or Nickel plated DZR sphere with full bore flow aperture. PTFE seats and stem seals, Extended Stem for Chilled Water Application, Lever operated, minimum Pressure rating, PN-20
 4. **Butterfly Valve -- Lever operated (65 to 150 mm):** Ductile Iron Butterfly Valves to BS EN 593, Semi-lugged / Fully- lugged, Ductile Iron Body (Epoxy Coated), Aluminum Bronze Disc, EPDM liner, Stainless Steel Shaft, Lever operated with long body neck, Pressure Rating, PN16 or PN 25 as per the system pressure requirements.
 5. **Butterfly Valve -- Electrically Gear operated (200 to 600 mm):** Ductile Iron Butterfly Valves to BS EN 593, Semi- lugged wafer pattern / fully lugged, Ductile Iron Body, Aluminum Bronze Disc, EPDM liner, Stainless Steel Shaft, Gear operated with Hand wheel, Pressure Rating, PN16 or PN 25 as per the system pressure requirements. High performance double eccentric type.
 6. **Balancing Valve – (15 mm to 50 mm):** Bronze Double Regulating valve, conforms to BS 7350, Threaded ends to BS 21, micro meter style hand wheel, bronze or DZR copper alloy body, DZR copper alloy trim, PN25 rated; fitted with pressure test points.
 7. **Balancing Valve – (65 mm & above):** Fixed Orifice Double Regulating Valves to BS7350 Double Regulating Valve Globe pattern with characterized throttling disc. Ductile Iron body, DZR trim, PTFE Disc Insert, Fitted with P84 insertion Pressure Test Points, Flanged Pressure Rating: PN16 or PN 25 as per the system pressure requirements.
 8. **Strainers: (15 to 50 mm):** Bronze Y-Strainers, Stainless Steel Screen (0.75 mm perforation), to be fitted with drain nipple, Screwed-in Cap, Threaded BS21 taper, Pressure Rating: PN20, WRAS approved.
 9. **Strainers: (65 to 600 mm):** Cast Iron Y-Strainers, Stainless Steel Screen (1.5mm perforation), Bolted Cap, Flanged ends to PN16, 25, Bosses drilled, tapped & plugged, Cap drilled, tapped & plugged to accept blow down valve, Pressure Rating, PN16 or 25.
 10. **Non-Return Valves (15 to 50 mm):** Bronze Swing Check Valves to BS EN 12288:2003 Series B, Metal Disc, Screwed-in Cap, Threaded BS21 Taper, BSI Kitemark approved, Pressure rating, PN25
 11. **Non-Return Valves (65 to 300 mm):** Cast Iron Check Valves, Wafer pattern, Cast Iron Body, Bronze trim, Resilient Seated (EPDM Body Seat Ring), Stainless Steel Spring, Suitable for mounting between PN16/ANSI 150 Flanges, Pressure rating, PN16.
 12. **Check Valves (65 to 300 mm):** Cast Iron Check Valves to BS EN 1234, Swing pattern, Cast Iron Body and Cover, Bronze trim, Bolted Cover, Flanged ends to PN16, Pressure Rating, PN16.
- F. Valve Sizes: Same as upstream piping unless otherwise indicated.
- G. Valve Actuator Types:
1. Gear Actuator: For quarter-turn valves NPS 8 (DN 200) and larger.

2. Hand wheel: For valves other than quarter-turn types.
 3. Hand lever: For quarter-turn valves NPS 6 (DN 150) and smaller, **except plug valves**.
 4. Wrench: For plug valves with square heads. Furnish Owner with 1 wrench for every 5 plug valves, for each size square plug-valve head.
 5. Chain wheel: Device for attachment to valve hand wheel, stem, or other actuator; of size and with chain for mounting height, as indicated in the "Valve Installation" Article.
- H. Valves in Insulated Piping: With 2-inch (50-mm) stem extensions and the following features:
1. Gate Valves: With rising stem.
 2. Ball Valves: With extended operating handle of non-thermal-conductive material, and protective sleeve that allows operation of valve without breaking the vapor seal or disturbing insulation.
 3. Butterfly Valves: With extended neck.
- I. Valve-End Connections:
1. Flanged: With flanges according to ASME B16.1 for iron valves.
 2. Solder Joint: With sockets according to ASME B16.18.
 3. Threaded: With threads according to ASME B1.20.1.
- J. Valve Bypass and Drain Connections: MSS SP-45.

2.2 BRONZE ANGLE VALVES

- A. Class 125/150/200/300. Bronze Angle Valves with Bronze Disc:
1. Manufacturers: Subject to compliance with requirements, provide from the list of approved manufacturers.
 2. Description:
 - a. Standard: MSS SP-80, Type 1.
 - b. CWP Rating: as required.
 - c. Body Material: ASTM B 62, bronze with integral seat and screw-in bonnet.
 - d. Ends: Threaded.
 - e. Stem and Disc: Bronze.
 - f. Packing: Asbestos free.
 - g. Hand wheel: Bronze

2.3 BRONZE BALL VALVES

- A. Class 150/200/300 Two-Piece, Full-Port, Bronze Ball Valves with Bronze or Stainless-Steel Trim:
1. Manufacturers: Subject to compliance with requirements, provide from the list of approved manufacturers.
 2. Description:
 - a. Standard: MSS SP-110.
 - b. CWP Rating: 4140 kPa.
 - c. Body Design: Two pieces.

- d. Body Material: Bronze.
- e. Ends: Threaded.
- f. Seats: PTFE or TFE.
- g. Stem: Bronze or Stainless steel.
- h. Ball: Chrome-plated brass or Stainless steel.
- i. Port: Full.

2.4 IRON, SINGLE-FLANGE BUTTERFLY VALVES

A. 200 CWP, Iron, Single-Flange Butterfly Valves with EPDM Seat and Ductile-Iron Disc:

- 1. Manufacturers: Subject to compliance with requirements, provide from the list of approved manufacturers.
- 2. Description:
 - a. Standard: MSS SP-67, Type I.
 - b. CWP Rating: 200 psig (1380 kPa) or higher as required.
 - c. Body Design: Lug type; suitable for bidirectional dead-end service at rated pressure without use of downstream flange.
 - d. Body Material: ASTM A 126, cast iron or ASTM A 536, ductile iron.
 - e. Seat: EPDM.
 - f. Stem: One- or two-piece stainless steel.
 - g. Disc: Nickel-plated ductile iron.

B. 200 CWP, Iron, Single-Flange Butterfly Valves with NBR Seat and Ductile-Iron Disc:

- 1. Manufacturers: Subject to compliance with requirements, provide from approved list of Manufacturer's.
- 2. Description:
 - a. Standard: MSS SP-67, Type I.
 - b. CWP Rating: 200 psig (1380 kPa) or higher as required.
 - c. Body Design: Lug type; suitable for bidirectional dead-end service at rated pressure without use of downstream flange.
 - d. Body Material: ASTM A 126, cast iron or ASTM A 536, ductile iron.
 - e. Seat: NBR.
 - f. Stem: One- or two-piece stainless steel.
 - g. Disc: Nickel-plated ductile iron.

2.5 IRON, GROOVED-END BUTTERFLY VALVES

A. 175/300 CWP, Iron, Grooved-End Butterfly Valves:

- 1. Manufacturers: Subject to compliance with requirements, provide from approved list of Manufacturer's.
- 2. Description:
 - a. Standard: MSS SP-67, Type I.
 - b. CWP Rating: 175 psig (1200 kPa) or higher as required.

- c. Body Material: Coated, ductile iron.
- d. Stem: One- or two-piece stainless steel.
- e. Disc: Coated, ductile iron.
- f. Seal: EPDM.

2.6 BRONZE SWING CHECK VALVES

- A. Class 150/200/300, Bronze Swing Check Valves with Bronze, Nonmetallic Disc:
 - 1. Manufacturers: Subject to compliance with requirements, provide from approved list of Manufacturer's.
 - 2. Description:
 - a. Standard: MSS SP-80, Type 4.
 - b. CWP Rating: 1035 kPa or higher as required.
 - c. Body Design: Horizontal flow.
 - d. Body Material: ASTM B 62, bronze.
 - e. Ends: Threaded.
 - f. Disc: Bronze, PTFE or TFE.

2.7 IRON FLANGED/GROOVED SWING CHECK VALVES

- A. Class 150/200/300, Iron, Dual-Plate Check Valves with Resilient Seat:
 - 1. Manufacturers: Subject to compliance with requirements, provide from approved list of Manufacturer's.
 - 2. Description:
 - a. Standard: API 594.
 - b. DN 65 to DN 300, CWP Rating: as required
 - c. DN 350 to DN 600, CWP Rating: as required.
 - d. Body Design: Clear or full waterway, Wafer, spring-loaded plates.
 - e. Body Material: ASTM A 126, gray iron, ASTM A 536, ductile iron.

2.8 BRONZE GATE VALVES

- A. Class 150/200/300, NRS Bronze Gate Valves:
 - 1. Manufacturers: Subject to compliance with requirements, provide from approved list of Manufacturer's.
 - 2. Description:
 - a. Standard: MSS SP-80, Type 1.
 - b. CWP Rating: as required.
 - c. Body Material: ASTM B 62, bronze with integral seat and union-ring bonnet.
 - d. Ends: Threaded.
 - e. Stem: Bronze.
 - f. Disc: Solid wedge; bronze.
 - g. Packing: Asbestos free.

- c. Hand wheel: bronze.

2.9 IRON GATE VALVES

- A. Class 150/200/300, NRS, Iron Gate Valves:
 - 1. Manufacturers: Subject to compliance with requirements, provide from approved list of Manufacturer's.
 - 2. Description:
 - a. Standard: MSS SP-70, Type I.
 - b. DN 65 to DN 300, CWP Rating: as required.
 - c. DN 350 to DN 600, CWP Rating: as required.
 - d. Body Material: ASTM A 126, gray iron with bolted bonnet.
 - e. Ends: Flanged.
 - f. Trim: Bronze.
 - g. Disc: Solid wedge.
 - h. Packing and Gasket: Asbestos free.

2.10 BRONZE GLOBE VALVES

- A. Class 150/200/300, Iron Globe Valves:
 - 1. Manufacturers: Subject to compliance with requirements, provide from approved list of Manufacturer's.
 - 2. Description:
 - a. Standard: MSS SP-85, Type I.
 - b. CWP Rating: as required.
 - c. Body Material: ASTM A 126, gray iron with bolted bonnet.
 - d. Ends: Flanged.
 - e. Trim: Bronze.
 - f. Packing and Gasket: Asbestos free.

PART 3 - EXECUTION

3.1 EXAMINATION

- A. Examine valve interior for cleanliness, freedom from foreign matter, and corrosion. Remove special packing materials, such as blocks, used to prevent disc movement during shipping and handling.
- B. Operate valves in positions from fully open to fully closed. Examine guides and seats made accessible by such operations.
- C. Examine threads on valve and mating pipe for form and cleanliness.

- D. Examine mating flange faces for conditions that might cause leakage. Check bolting for proper size, length, and material. Verify that gasket is of proper size, that its material composition is suitable for service, and that it is free from defects and damage.
- E. Do not attempt to repair defective valves; replace with new valves.

3.2 VALVE INSTALLATION

- A. Install valves with unions or flanges at each piece of equipment arranged to allow service, maintenance, and equipment removal without system shutdown.
- B. Locate valves for easy access and provide separate support where necessary.
- C. Install valves in horizontal piping with stem at or above center of pipe.
- D. Install valves in position to allow full stem movement.
- E. Install chain wheels on operators for valves DN 100 and larger and more than 2400 mm above floor. Extend chains to 1520 mm above finished floor.
- F. Install check valves for proper direction of flow and as follows:
 - 1. Swing Check Valves: In horizontal position with hinge pin level.

3.3 ADJUSTING

- A. Adjust or replace valve packing after piping systems have been tested and put into service but before final adjusting and balancing. Replace valves if persistent leaking occurs.

3.4 GENERAL REQUIREMENTS FOR VALVE APPLICATIONS

- A. If valve applications are not indicated, use the following:
 - 1. Shutoff Service: Ball, butterfly or gate valves.
 - 2. Butterfly Valve Dead-End Service: Single-flange lug type.
 - 3. Throttling Service: Globe valves.
 - 4. Pump-Discharge Check Valves:
 - a. DN 50 and Smaller: Bronze swing check valves with bronze disc.
 - b. DN 65 and Larger: Iron swing check valves
- B. If valves with specified SWP classes or CWP ratings are not available, the same types of valves with higher SWP classes or CWP ratings may be substituted.
- C. Select valves, except wafer types, with the following end connections:
 - 1. Pipes DN 50 and Smaller: Threaded ends
 - 2. Pipes DN 65 to DN 100: Flanged ends
 - 3. Pipes DN 125 and Larger: Flanged ends.

3.5 DOMESTIC, HOT- AND COLD-WATER VALVE SCHEDULE (As Applicable)**A. Pipe NPS 2 (DN 50) and Smaller:**

1. Bronze Valves: May be provided with solder-joint ends instead of threaded ends.
2. Bronze Angle Valves: Class 125 bronze disc.
3. Ball Valves: One piece, regular port, bronze with bronze trim.
4. Bronze Swing Check Valves: Class 125 or Class 150 as applicable to system pressure requirement bronze disc.
5. Bronze Gate Valves: Class 125 or Class 150 as applicable to system pressure requirement, NRS.
6. Bronze Globe Valves: Class 125 or Class 150 as applicable to system pressure requirement, bronze disc.

B. Pipe NPS 2-1/2 (DN 65) and Larger:

1. Iron Valves, NPS 2-1/2 to NPS 4 (DN 65 to DN 100): May be provided with threaded ends instead of flanged ends.
2. Iron Ball Valves: Class 150.
3. Iron, Single-Flange Butterfly Valves: 200 CWP, EPDM seat, ductile-iron disc.
4. Iron, Grooved-End Butterfly Valves: 175 or 300 CWP, as applicable to system pressure requirement.
5. Iron Swing Check Valves: Class 125 or Class 250 as applicable to system pressure requirement, metal seats.
6. Iron Swing Check Valves with Closure Control: Class 125, lever and spring or weight as required.
7. Iron, Grooved-End Swing Check Valves: 300 CWP.
8. Iron Gate Valves: Class 125 or Class 150 as applicable to system pressure requirement NRS.
9. Iron Globe Valves: Class 125 or Class 150 as applicable to system pressure requirement.

3.6 SANITARY-WASTE AND STORM-DRAINAGE VALVE SCHEDULE (As Applicable)**A. Pipe NPS 2 (DN 50) and Smaller:**

1. Bronze Valves: May be provided with solder-joint ends instead of threaded ends.
2. Bronze Angle Valves: Class 125 or Class 150 as applicable to system pressure requirement, bronze disc.
3. Ball Valves: One-piece, regular port, bronze with bronze trim.
4. Bronze Swing Check Valves: Class 125 or Class 150 as applicable to system pressure requirement, bronze disc.
5. Bronze Gate Valves: Class 125 or Class 150 as applicable to system pressure requirement, NRS.
6. Bronze Globe Valves: Class 125 or Class 150 as applicable to system pressure requirement, bronze disc.

B. Pipe NPS 2-1/2 (DN 65) and Larger:

1. Iron Valves, NPS 2-1/2 to NPS 4 (DN 65 to DN 100): May be provided with threaded ends instead of flanged ends.
2. Iron Ball Valves: Class 150.

3. Iron Swing Check Valves: Class 125 or Class 250 as applicable to system pressure requirement, metal seats.
4. Iron Swing Check Valves with Closure Control: Class 125, lever and weight.
5. Iron, Grooved-End Swing Check Valves: 300 CWP.
6. Iron Gate Valves: Class 125 or Class 250 as applicable to system pressure requirement, NRS.
7. Iron Globe Valves: Class 125 or Class 150 as applicable to system pressure requirement,

END OF SECTION 220523